

Amazon Music Song Clustering Using Unsupervised Machine Learning

1. Project Overview

The objective of this project is to group similar songs based on their audio characteristics using Unsupervised Machine Learning techniques. Music streaming platforms contain thousands of songs, making it difficult to manually categorize tracks according to their sound patterns.

This project utilizes clustering algorithms to identify hidden relationships among songs and automatically organize them into meaningful groups. The project includes data preprocessing, feature scaling, cluster analysis, evaluation, visualization, and deployment through an interactive Streamlit dashboard.

2. Problem Statement

Music recommendation systems require an understanding of similarities between songs. Traditional categorization based solely on genres may not accurately represent the actual characteristics of a song.

The goal of this project is to use audio features such as danceability, energy, loudness, tempo, and acousticness to automatically cluster songs with similar musical properties.

3. Dataset Description

The dataset contains Amazon Music songs along with their audio characteristics.

Features Used

- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence
- Tempo
- Duration_ms

Reference Features

- Song Name
- Artist Name
- Genre

Target Variable

No target variable is used because clustering is an Unsupervised Learning problem.

4. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset into a Pandas DataFrame.
2. Checked dataset structure, data types, and summary statistics.
3. Identified missing values and duplicate records.
4. Selected relevant numerical audio features for clustering.
5. Standardized feature values using StandardScaler.
6. Prepared the final feature matrix for clustering.

After preprocessing, the dataset was ready for cluster analysis.

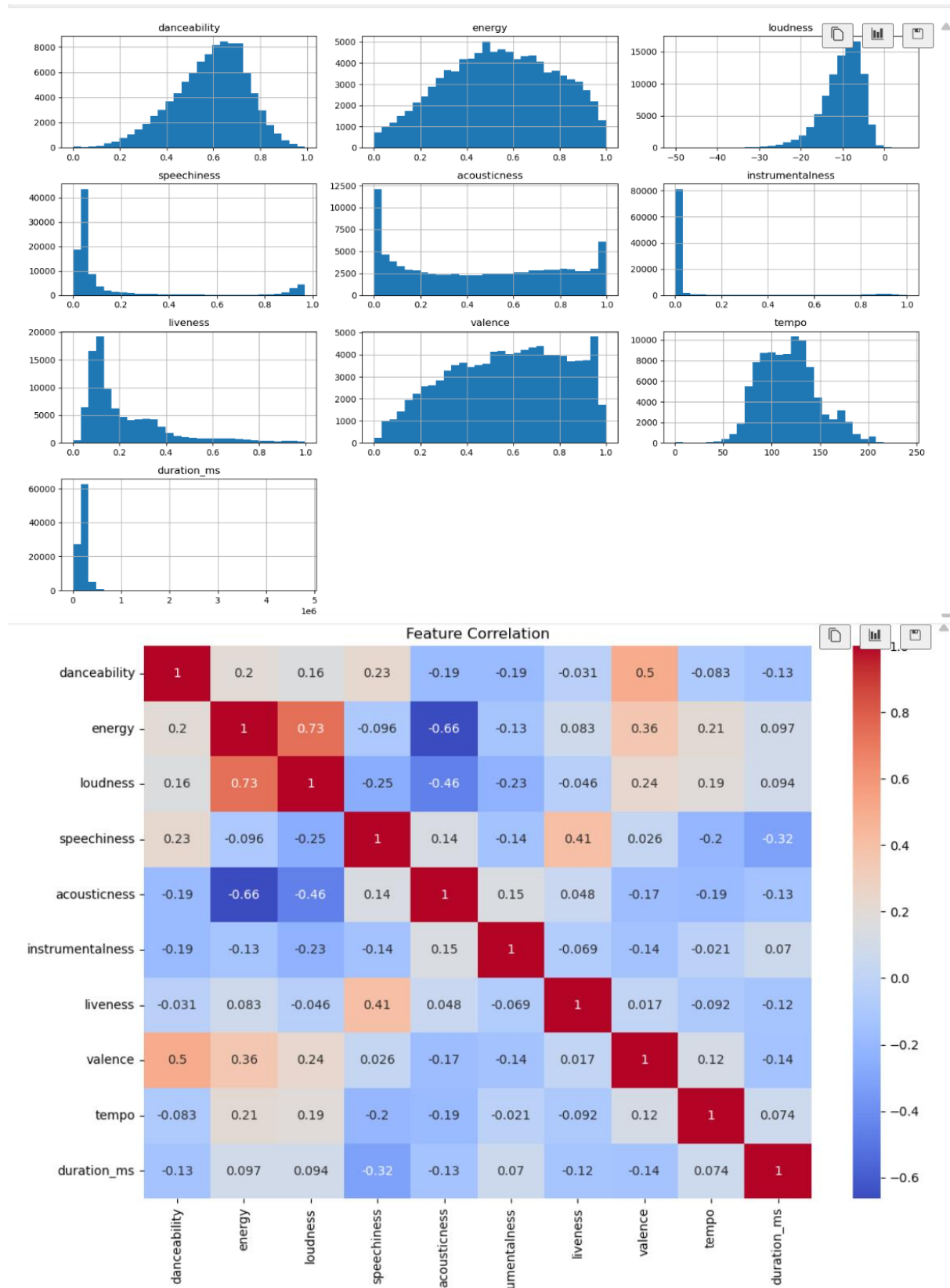
5. Feature Selection

The following audio features were selected because they describe the rhythm, mood, intensity, and instrumentation of a song:

- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence

- Tempo
- Duration_ms

These features provide a comprehensive representation of a song's sound profile.



6. Machine Learning Models Implemented

Three clustering algorithms were explored and evaluated.

1. K-Means Clustering

A centroid-based clustering algorithm that partitions songs into groups based on similarity.

2. DBSCAN

A density-based clustering algorithm capable of identifying outliers and irregular cluster shapes.

3. Hierarchical Clustering

An algorithm that builds clusters by progressively merging similar songs into larger groups.

7. Determining Optimal Clusters

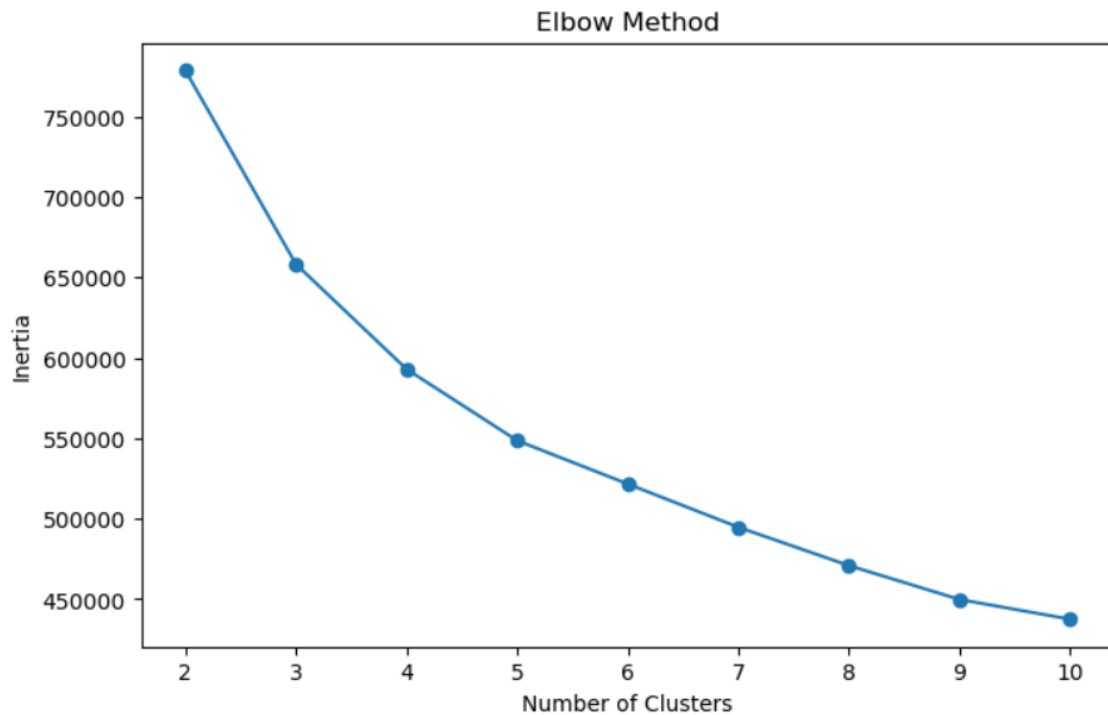
Elbow Method

The Elbow Method was used to determine the optimal number of clusters by analyzing inertia values across different cluster counts.

Silhouette Score

Silhouette Score was used to measure how well songs fit within their assigned clusters.

The optimal cluster count was selected based on both methods.



K=2: 0.1998
K=3: 0.2378
K=4: 0.2273
K=5: 0.1857
K=6: 0.1872
K=7: 0.1648
K=8: 0.1721
K=9: 0.1726
K=10: 0.1579

8. Cluster Evaluation Metrics

The following evaluation metrics were used:

Silhouette Score

Measures how similar a song is to its own cluster compared to neighboring clusters.

Davies-Bouldin Index

Measures cluster compactness and separation.

Inertia

Measures the compactness of clusters and is used in the Elbow Method.

9. Cluster Analysis

The clustering algorithm successfully grouped songs into distinct clusters based on their audio characteristics.

Example Cluster Interpretations

Cluster 0

- High Energy
- High Danceability
- Suitable for party playlists

Cluster 1

- High Acousticness
- Low Energy
- Relaxing and acoustic tracks

Cluster 2

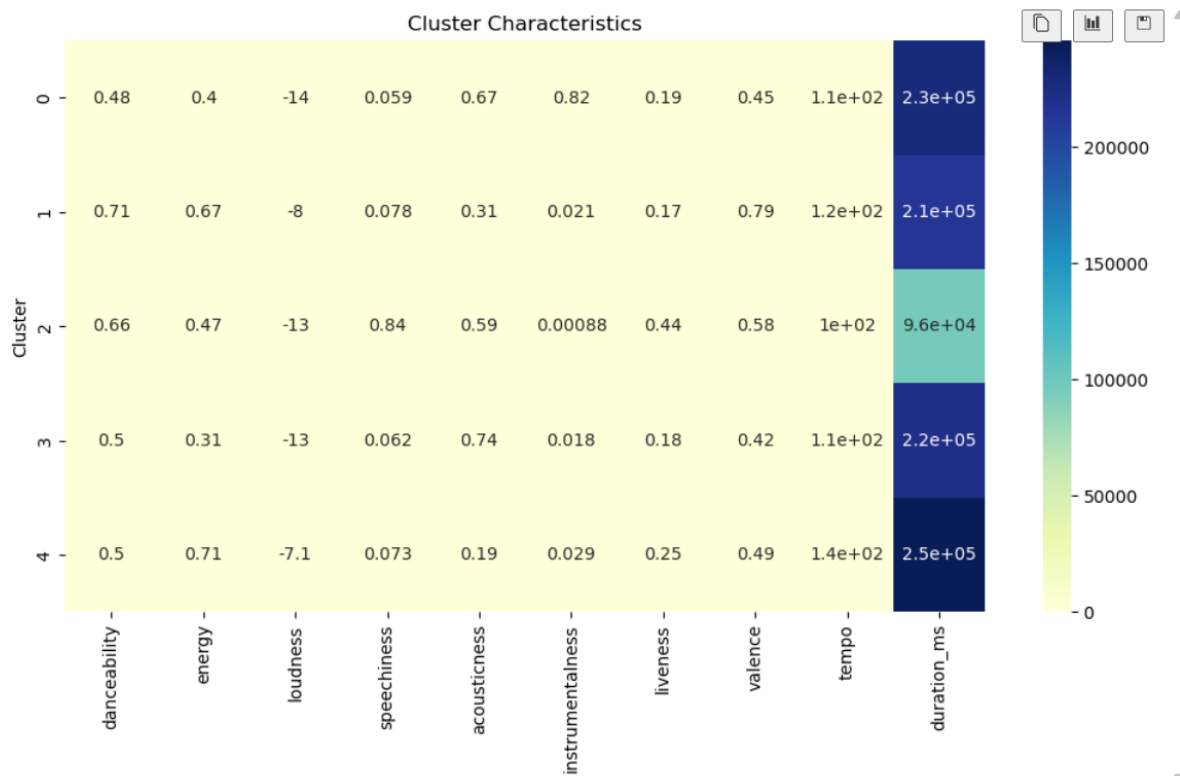
- High Instrumentalness
- Minimal vocals
- Instrumental music

Cluster 3

- Moderate Energy
- Balanced musical characteristics
- General listening songs

Cluster 4

- High Tempo
- High Valence
- Upbeat and positive tracks



10. PCA Visualization

Principal Component Analysis (PCA) was used to reduce the dataset into two dimensions for visualization.

Benefits:

- Easier cluster interpretation
- Visual understanding of song groups
- Identification of cluster separation

The PCA scatter plot demonstrated clear grouping of songs into distinct clusters.



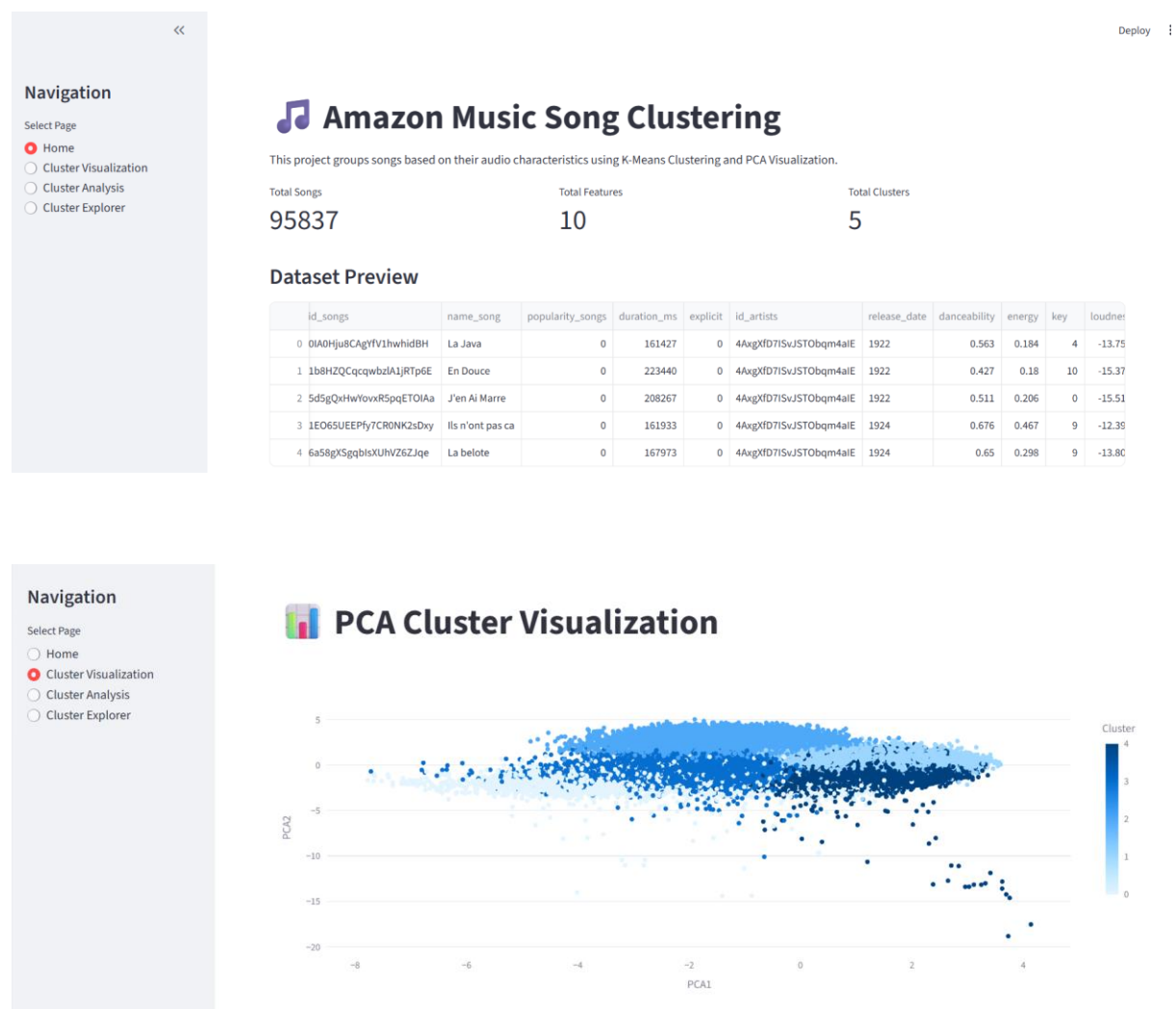
11. Streamlit Application Development

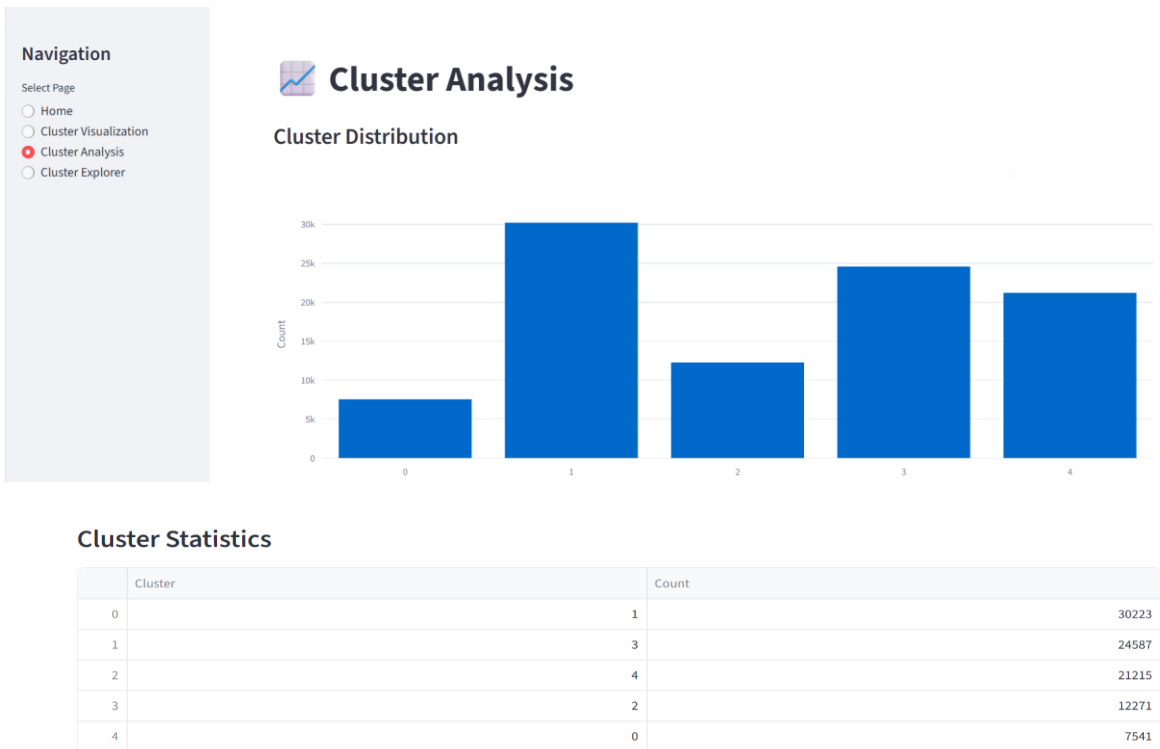
A Streamlit web application was developed to provide an interactive visualization platform.

Features of the Application

- User-friendly interface
- PCA visualization
- Cluster distribution charts
- Interactive data tables
- Real-time cluster exploration

The application loads the clustered dataset and visualizes clustering results dynamically.





Cluster Explorer

Select Cluster

0

Songs in Cluster 0: 7541

	name_song	name_artists	Cluster
53	Veux Tu Que J'te Fasse Un Dessin	Georgius	0
65	West End Blues	Louis Armstrong & His Hot Five	0
67	Beau Koo Jack	Louis Armstrong & His Hot Five	0
69	Gut Bucket Blues	Louis Armstrong & His Hot Five	0
70	Skid-Dat-De-Dat	Louis Armstrong & His Hot Five	0
72	Skip The Gutter	Louis Armstrong & His Hot Five	0
74	Cornet Chop Suey	Louis Armstrong & His Hot Five	0
78	Savoy Blues	Louis Armstrong & His Hot Five	0
82	Come Back, Sweet Papa	Louis Armstrong & His Hot Five	0
84	Who'sit	Louis Armstrong & His Hot Five	0

12. Project Structure

AMAZON MUSIC CLUSTERING PROJECT/

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| — Report/

| | — Amazon Music Song Clustering Report.pdf

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|
|— Screenshots/
|   |— Home Page.png
|   |— PCA Visualization.png
|   |— Cluster Analysis.png
|   └─ Cluster Explorer.png
|
|— app.py
|— genre_artists_analysis.ipynb
|— README.md
|— requirements.txt
|— scaler.pkl
|— kmeans_model.pkl
|— pca.pkl
|— songs_clustered.csv
└─ single_genre_artists.csv
```

13. Results

The project successfully achieved the following objectives:

- Loaded and analyzed the Amazon Music dataset.
 - Performed data preprocessing and feature scaling.
 - Determined the optimal number of clusters.
 - Applied K-Means clustering successfully.
 - Evaluated clustering quality using multiple metrics.
 - Visualized clusters using PCA.
 - Developed an interactive Streamlit dashboard.
 - Generated meaningful song groupings for recommendation purposes.
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14. Conclusion

This project demonstrates the successful application of Unsupervised Machine Learning techniques for clustering Amazon Music songs based on their audio characteristics.

K-Means clustering effectively grouped similar songs into meaningful clusters. PCA visualization provided insights into cluster distribution, while evaluation metrics confirmed the quality of clustering results.

The developed Streamlit application allows users to explore clusters interactively and understand relationships among songs. The project can serve as a foundation for music recommendation systems, playlist generation, and music discovery platforms.

15. Future Scope

1. Implement advanced clustering algorithms.
2. Integrate recommendation engine functionality.
3. Use deep learning-based music embeddings.
4. Deploy the application on cloud platforms.
5. Build personalized playlist generation systems.
6. Integrate real-time music streaming data.
7. Develop automated music recommendation solutions for streaming platforms.